VEX Rover Mechanisms Challenge

Unit 1 Project Brief

Engineering Scenario

Planetary rovers need mechanisms that can lift, deploy, steer, transfer, clamp, or position parts reliably while using limited power and mass.

Design Goal

Design, build, and test a VEX mechanism that redirects force, speed, distance, or motion for a rover mission-support task.

Criteria	• Mechanism performs a clear rover-related task. • Team explains mechanical advantage, energy transfer, or power trade-offs. • Prototype performance is measured with repeated tests. • Final recommendation is based on data and observed limitations.
Constraints	• Use VEX parts and approved fasteners only unless the teacher approves an add-on. • Prototype must be safe to operate on the tabletop or floor test area. • Testing must include at least three repeated trials.
Required Evidence	• Problem statement and mechanism concept sketches • System diagram showing input, process, and output • Mechanical calculation or trade-off explanation • Test data and revision notes • Final design review summary
Checkpoints	• Define the mechanism task • Generate team concepts • Select and build one concept • Test and revise • Present the engineering recommendation

Final Design Review Expectations

Your final review should clearly connect the problem, evidence, prototype decisions, testing results, limitations, and next recommended improvement. Strong teams show how data changed the design rather than only describing what they built.

Student Deliverable Reminder

Keep all sketches, calculations, data tables, photos, revision notes, and decisions in your engineering notebook or assigned project packet. The notebook should make your design process visible.